Transportation Research Part C 191 (2026) 105818



Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc





Aligning LLM agents with human learning and adjustment behavior: A dual agent approach

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ARTICLE INFO

Keywords:
Travel demand modeling
Large language model agent
Day-to-day behavior
Automatic calibration
Agent-based simulation

ABSTRACT

Effective modeling of how human travelers learn and adjust their travel behavior from interacting with transportation systems is critical for system assessment and planning. However, this task is also difficult due to the complex cognition and decision-making involved in such behavior. Recent research has begun to leverage Large Language Model (LLM) agents for this task. Building on this, we introduce a novel dual-agent framework that enables continuous learning and alignment between LLM agents and human travelers in learning and adaptation behavior from online data streams. Our approach involves a set of LLM traveler agents, equipped with a memory system and a learnable persona, which serve as simulators for human travelers. To ensure behavioral alignment, we introduce an LLM calibration agent that leverages the reasoning and analytical capabilities of LLMs to train the personas of these traveler agents. Working together, this dual-agent system is designed to track and align the underlying decision-making mechanisms of travelers and produce realistic, adaptive simulations. Using a real-world dataset from a day-to-day route choice experiment, we show our approach significantly outperforms existing methods in individual behavior alignment and LLM-based methods in aggregate flow prediction performance. Furthermore, we demonstrate that our method moves beyond simple behavioral mimicry to capture the evolution of underlying learning processes, which contributes to robust generalization. Overall, our framework provides a new approach for creating adaptive and behaviorally realistic agents to simulate travelers' learning and adaptation that can benefit transportation simulation and policy analysis.

1. Introduction

Agent-based modeling has become a pivotal paradigm in transportation system analysis and planning (Kagho et al., 2020). This approach simulates system-level dynamics by modeling the interactions of autonomous agents, representing individual travelers, within a virtual environment. The performance of the underlying mobility infrastructure is determined by a traffic simulator in response to the agents' collective behavior. Crucially, this interactive process is designed to emulate human learning and adjustment: agents repeatedly engage with the infrastructure, receive performance feedback such as travel times and costs, and iteratively refine their travel plans and strategies. Therefore, the ability of agents to accurately model the learning and adjustment behavior of their

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https://doi.org/10.1016/j.trc.2026.105818
Received 31 October 2025; Received in revised form 12 April 2026; Accepted 15 June 2026
Available online 23 June 2026
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human counterparts is critical for an agent-based model to correctly characterize transportation system dynamics and inform sound planning decisions. A mischaracterization of this behavior risks causing analysts to overlook emergent dynamics and can lead the simulation to converge on incorrect stationary points, thereby compromising the entire downstream planning process. Specifically, this can lead to misguided infrastructure investments based on inaccurate cost-benefit analyses or the rejection of effective policies due to a fundamental misunderstanding of public adaptation over time.

However, modeling the learning and adjustment behavior of travelers is a challenging task due to the complex cognitive process behind this behavior, beyond the impacts of the current system state and rationality. Indeed, research has shown that travelers' learning and decision-making are greatly impacted by their past travel experience and bounded rationality. On the impact of experience, research has shown that factors such as the utility of past travel experiences (Mahmassani, 2009; Iida et al., 1992; Meneguzzer and Olivieri, 2013), travel time variability (Avineri and Prashker, 2005), significant positive or negative experiences (Sönmez and Graefe, 1998), and exploration (Lowry, 2024) have significant impacts on how travelers make travel-related decisions, and it has also been shown that inclusion of past experiences can improve the modeling accuracy (Bamberg et al., 2003). On the bounded rationality of travelers, existing studies have also found evidence of multiple types of bounded rational behavior in travelers' learning and adjustment process, including inertia (Qi et al., 2023), habit (Schlich and Axhausen, 2003), indifference between similar travel times (Zhao and Huang, 2016), and risk aversion/seeking (Avineri and Prashker, 2005; Schwanen and Ettema, 2009; Zhang et al., 2018). Overall, the complex interplay between traveler characteristics, past experience, and bounded rationality makes modeling human learning and adjustment challenging, especially for mathematical models that generally require *a priori* assumptions. Furthermore, existing models such as day-to-day dynamic systems can simulate complex population-level outcomes but lack the resolution to analyze individual travelers, as they do not account for modeling individuals' decision-making processes (Jin, 2020; Zhou and Mahmassani, 2007; Yu et al., 2020).

To address these limitations, researchers have recently proposed leveraging Large Language Models (LLMs) to advance agent-based modeling in transportation (Liu et al., 2025c; Adornetto et al., 2025; Song et al., 2025; Liu et al., 2025a). LLMs, such as GPT (Achiam et al., 2023), Gemini (Team et al., 2023), and Llama (Touvron et al., 2023), are a class of generative artificial intelligence renowned for producing high-quality, human-like text from natural language instructions. This impressive general intelligence has prompted researchers to propose utilizing LLM agents, which are autonomous agents whose decisions are powered by LLMs, as the core of agent-based modeling frameworks. Compared to established models, LLMs offer distinct potential advantages. Their training on vast datasets of human-generated text, combined with their emergent capabilities for instruction-following (Wei et al., 2021) and zero-shot learning (Kojima et al., 2022), provides the potential to simulate complex human behaviors with fewer a priori assumptions (Mo et al., 2023; Liu et al., 2024, 2025c; Nie et al., 2025). Furthermore, their inherent ability to process and generate natural language simplifies the development and application of these agents, making them more accessible compared to complex mathematical models (Liu et al., 2025c).

Despite the potential of LLMs, realistically modeling the dynamic learning and adjustment processes of travelers remains a formidable challenge. Direct application of base LLMs is suboptimal, as recent studies have documented significant behavioral dissimilarities between them and human travelers (Wang et al., 2025b). Consequently, methods are needed to condition LLMs into behaviorally plausible agents. However, research addressing this challenge is scarce, and existing methods invariably rely on pre-specified behavioral assumptions (Wang et al., 2025a). A clear research gap therefore exists for methods that can leverage the full expressive power of LLMs to create behaviorally accurate simulations of learning and adjustment, which our study aims to fill.

In this study, we propose a dual-LLM agent framework designed to automatically align the learning and adjustment behaviors of simulated agents with human travelers. At the center of our approach are two sets of LLM agents: (1) a series of LLM traveler agents, each of which simulates daily travel decisions based on a dynamic memory and a core persona, and (2) an LLM calibration agent, which leverages an LLM's reasoning to optimize the personas of the traveler agents against data streams of human travel behavior. Methodologically, our calibration agent is driven by a textual 'pseudo-gradient' descent for efficient refinement, and is further regularized by a smoothing mechanism to enhance generalization. Using a real-world dataset from a human day-to-day route choice experiment (Wang et al., 2025b), we assess our approach and benchmark its performance against existing LLM agent baselines and representative machine-learning-based approaches. The results show that our approach can achieve more accurate alignment of travelers' day-to-day learning behavior than existing approaches, and achieve better aggregate flow prediction performance than existing LLM agents. Furthermore, the evaluation results also reveal our method's ability to extend beyond replicating behavioral outcomes to aligning the underlying tendency and evolution of learning and decision-making. This enables our agent to capture how these strategies evolve over time, leading to robust generalization.

To our knowledge, our work presents the first framework for automatically calibrating LLM agents to simulate the dynamic learning and adjustment of travelers over time. Methodologically, we introduce a novel dual-agent architecture where an LLM calibration agent supervises the calibration of an LLM traveler simulation agent with online updates. This allows our system to capture the evolution of travel behavior and leverage the expressiveness of natural language to capture the complex behavior of human travelers. Practically, this framework enables researchers to leverage standard, API-based LLMs to create adaptive and realistic agents to inform agent-based simulations. Our approach contributes to the emerging literature on building more behaviorally realistic and longitudinally valid LLM-based tools for transportation modeling.

The remainder of this paper is structured as follows: Section 2 provides a review of relevant literature on LLM traveler agents and existing methodologies of aligning them with human travelers, and also further positions our work. Section 3 presents the mathematical settings of the problem and details our approach. Section 4 outlines our dataset, experiment setting, evaluation metrics, and compared approaches. The results and discussions of the experimental results are presented in Section 5. Finally, Section 6 concludes the paper.

2. Related work

Our work is closely related to the recent stream of research on utilizing LLMs for human travel behavior modeling and simulation. Below, we review the relevant existing works and further position the methodological contribution of our paper.

The advent of LLMs, with their sophisticated ability to interpret and generate natural language, has led researchers to explore their use as a "silicon sample" of human behavior (Grossmann et al., 2023; Hutson, 2023). In travel behavior research, LLMs are being developed to simulate travelers, offering a way to generate synthetic travelers and facilitate behavior and system modeling while relaxing the requirement of *a priori* assumptions of traditional approaches (Liu et al., 2025c; Nie et al., 2025). Consequently, much literature has emerged to validate the behavioral alignment of base LLMs with human travelers across a wide spectrum of decisions. This includes extensive research into travel mode choice (Mo et al., 2023; Liu et al., 2024; Nishida et al., 2025; Xu and Jiao, 2025), destination choice (Wang et al., 2023; Beneduce et al., 2025), travel attitudes (Tzachristas et al., 2025; Xu and Wang, 2025), the value of travel time (Yan et al., 2026), responses to policy changes (Yan et al., 2025), day-to-day learning (Wang et al., 2025b), and parking search behavior (Fulman et al., 2025).

Collectively, these studies confirm that LLMs can interpret travel scenarios and reason through decision-making processes in a human-like manner. They demonstrate an ability to leverage their internal knowledge base to generate plausible behaviors (Mo et al., 2023; Wang et al., 2023; Beneduce et al., 2025; Zhang and Xu, 2026; Yan et al., 2025) and even replicate human tendencies and biases in various contexts (Yan et al., 2026; Fulman et al., 2025). Despite these strengths, the literature also consistently identifies significant behavioral misalignments. Key limitations include discrepancies in preferences (Liu et al., 2024; Nishida et al., 2025) and values (Xu and Wang, 2025), reduced sensitivity and elasticity to contextual factors (Yan et al., 2026), and divergent decision-making rules compared to humans (Wang et al., 2025b). Overall, existing research forms a clear consensus: while LLMs hold immense potential for behavioral simulation, they require further conditioning to ensure their outputs are both accurate and reliable for research and application.

To address the challenges in conditioning LLMs for reliable behavioral simulation, recent research in travel behavior has increasingly focused on applying LLM agents, which are autonomous entities whose perception and decision-making processes are driven by an underlying LLM, to enhance alignment and facilitate behavioral simulation. Existing methodologies for developing these agents to accurately imitate human travel behavior primarily fall into two distinct categories,

- **Memory system enhancement:** This approach equips LLM agents with a structured memory module to store past experiences (actions, outcomes, reflections). During decision simulation, relevant memories are retrieved and incorporated into the LLM's prompt, typically via retrieval-augmented generation techniques. This strategy is effective for two key reasons: (1) It mirrors the cognitive process where humans leverage past experiences for learning and decision-making. (2) It capitalizes on the LLM's few-shot learning capabilities (Brown et al., 2020), allowing the model to treat retrieved memories as contextual examples, discern underlying behavioral patterns, and apply them to new situations. The utility of memory-augmented LLMs for behavioral simulation has been demonstrated across various transportation contexts (Liu et al., 2024; Alsaleh and Farooq, 2025; Xu and Jiao, 2025; Zhang and Xu, 2026; Xu and Wang, 2025). Currently, techniques on using them to enhance LLM traveler agents include utilizing memory systems to improve multi-agent simulation convergence (Liu et al., 2025b), guiding agent trajectory and activity planning through summarized experiences (Wang et al., 2024; Liu et al., 2025c), and enhancing the simulation of specific choices like public transit usage or activity locations (Ge et al., 2025; Chen et al., 2025).
- **Persona learning:** This strategy assigns each LLM agent a persona that explicitly describes the intrinsic values, preferences, and decision-making characteristics of the human traveler being simulated. Analogous to parameters in traditional utility functions, this persona provides the LLM with a consistent identity to adopt during generation, thereby improving behavioral alignment. The primary methodology involves inferring these personas inversely from empirical behavioral data. Research has consistently shown that employing such inferred personas significantly enhances the fidelity of LLM agent simulations. For example, Liu et al. (2024) first applied preference personas that are inversely learned from a stated preference dataset to assist behavior simulation, and concluded that this approach can greatly enhance such alignment. Subsequent work has further validated the effectiveness of learning personas from various data sources to better simulate choices related to activity planning and mode selection (Chen et al., 2025; Liu et al., 2026; Sameen et al., 2025).

To imitate human travelers' learning and adjustment behavior, it is necessary to provide LLM with the information on both the context (including past memories) and how the traveler makes decisions (included in the persona). Therefore, on aligning the learning and adjustment behavior, both the memory and the persona are necessary for the agents. In a closely related paper, Wang et al. (2025a) proposed giving the agent a pre-assigned persona that describes the Big-Five personality traits (Roccas et al., 2002) and a memory processing system of exponential smoothing to simulate travelers' day-to-day route choice behavior, and evaluated their alignment with human travelers. Compared to the aforementioned studies, our paper makes two main methodological contributions,

1. **We relax the assumptions and restrictions on the agent persona.** Compared with existing studies that either employ *a-priori* personas (Wang et al., 2025a) or a Likert scale measurement of pre-specified factors (Liu et al., 2024, 2026; Sameen et al., 2025), we relax the format restrictions of the pre-specified personas by reducing the formatting restrictions of persona descriptions, enabling the agents to better leverage natural language to describe complex human behavior.
2. **To support this relaxation, we introduce automated learning guided by another LLM agent for the traveler agents.** Relaxation of the persona format leads to a large feasible region, thereby challenging the persona learning procedure. Therefore, we introduce another LLM agent to automatically calibrate the agent persona, as LLM's language reasoning ability particularly suits

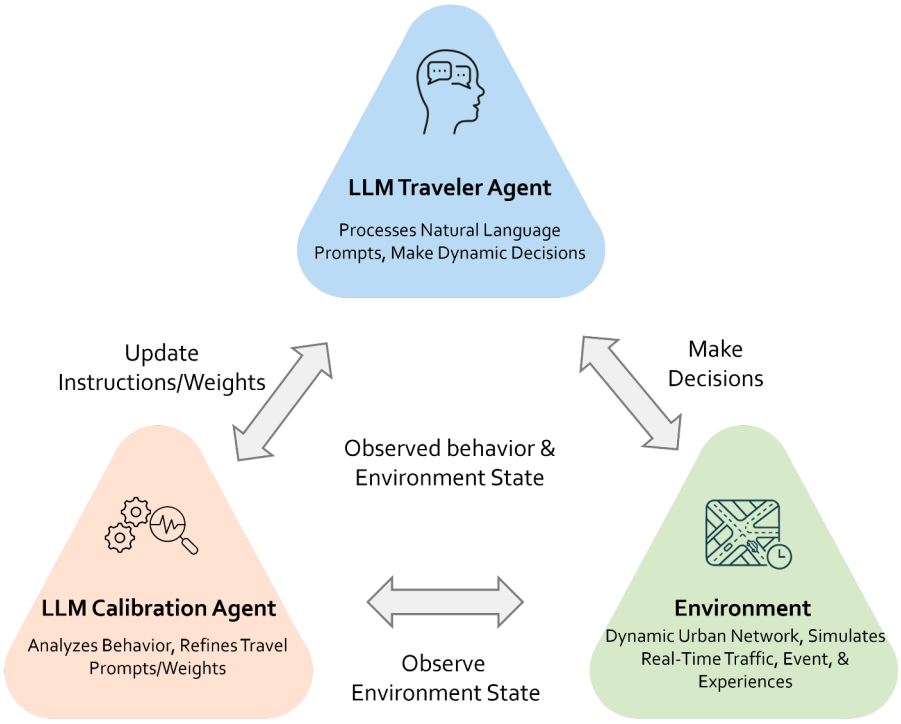


Fig. 1. Overview of our approach.

this task. This idea is similar to recent efforts to leverage the "LLM for Science" agents for automatic discovery in transportation (Lai et al., 2025; Guo et al., 2025), but ours is the first instance in using this approach to facilitate travel behavior alignment.

3. Methodology

In this section, we formally define the LLM agent alignment problem and outline our approach. First, we present the definition of the dynamics of the transportation system and outline our approach in Section 3.1. The design and training process of the LLM traveler agents and the LLM calibration agent are then presented in Sections 3.2 and 3.3, respectively.

3.1. Premise

To establish our framework, we first define the transportation system and its dynamics. The system consists of a set of M heterogeneous agents, $\mathcal{N} = \{n_1, n_2, \ldots, n_M\}$, interacting with an environment, $\mathcal{E}$. Each agent $n_i \in \mathcal{N}$ represents a traveler, while the environment $\mathcal{E}$ encompasses the transportation infrastructure. The interaction between agents and the environment occurs over a sequence of discrete time periods, indexed by $t = 1, 2, \ldots$. In each period t, the system evolves through the following sequential process,

1. Agent Decision-Making. In each period t, agent $n_i \in \mathcal{N}$ makes a travel decision, a_i^t. This decision is determined by the agent's decision-making function, f_i, which describes the agent's behavior of referring to their past experiences and making decisions based on their characteristics,

$$a_i^t = f_i\left(\{a_i^\tau, o_i^\tau\}_{\tau=1}^{t-1}\right) \tag{1}$$

in which $\{a_i^\tau, o_i^\tau\}_{\tau=1}^{t-1}$ is the collection of the agent's past decisions (a_i^τ) and their observed outcomes (o_i^τ).

2. Environment state update. Following the agents' collective decisions $\{a_i^t\}_{i=1}^M$ in period t, the traffic condition changes and thus the environment transitions to a new state, s^t. This process is governed by a known system dynamics function, E, which updates the state based on the previous state s^{t-1} and the agents' decisions,

$$s^t = E\left(\{a_i^t\}_{i=1}^M, s^{t-1}\right) \tag{2}$$

Here, the state vector s^t describes the performance of the transportation infrastructure (e.g., link travel times, traffic volumes). The function E represents the underlying model of the system, such as a set of link performance functions in a traffic assignment context. As is standard in agent-based modeling, the function E is specified by the modeler to represent the ground-truth dynamics of the simulated environment, using tools such as traffic simulators.

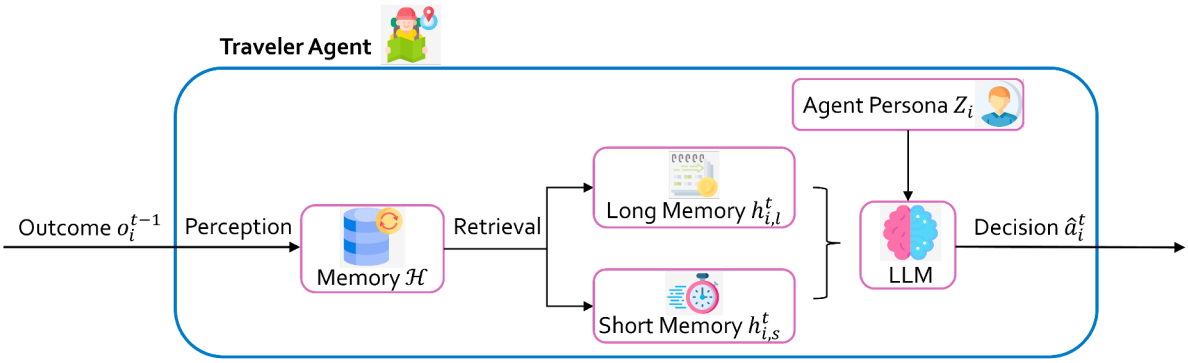


Fig. 2. Components and workflow of an LLM traveler agent.

3. Agent observation. After the environment transitions to the new state s^t, each agent n_i receives feedback based on their individual experience. Generally, in a transportation system, a traveler typically only knows a part of the system state, such as only knowing the travel time of their chosen route. To reflect this partial observability, we model the agent's personal observation, o_i^t, as a function of the global state and their specific decision,

$$o_i^t = O(s^t, a_i^t) \tag{3}$$

in which the observation function, O, maps the system state and the agent's decision to a personalized outcome. For example, O would take the full set of network travel times (s^t) and the agent's chosen route (a_i^t) to return the specific travel time they experienced. This new experience pair, (a_i^t, o_i^t), is then added to the agent's history, forming the basis for their decision in the next period.

As the interactive process defined by Eqs. (1)–(3) unfolds, the modeler continuously receives data on agent decisions, their observations, and the resulting system states. At each time period, the task for the modeler is to learn a set of surrogate decision-making models, $\{\hat{f}_i\}$, one for each agent. These surrogate models, $\hat{f}_i$, are designed to approximate the corresponding human traveler's true, unobserved behavioral strategies (represented by f_i in Eq. (1)). The goal is to use these learned models to generate simulations of the human agents' future actions (a_i^t) and the resulting aggregate system states. Therefore, the objective is to learn $\{\hat{f}_i\}$ so that each agent's future behavior is well-aligned with the behavior of the human it represents, and the modeler's goal can be conceptualized as finding a model $\hat{f}_i$ that minimizes the expected future behavior simulation error. Mathematically, at the end of time period T, the agent alignment optimization problem for the modeler is,

$$\min_{\hat{f}_i} \mathbb{E}_{t>T}\left[\rho\left(\hat{f}_i\left(\{a_i^\tau, o_i^\tau\}_{\tau=1}^{t-1}\right), a_i^t\right) \mid \{a_i^\tau, o_i^\tau\}_{\tau=1}^{T}\right] \tag{4}$$

where the expectation $\mathbb{E}$ is taken over all possible future trajectories given the history from time period 1 to T, and ρ is a loss function that quantifies the discrepancy between the model's prediction and the corresponding traveler's actual decision. By solving Eq. (4), the modeler aims to create the most robust and generalizable behavior simulator of what the agent will do next that can be applied in an agent-based simulation.

To solve the optimization problem 4, we propose a dual-LLM-agent approach that automatically learns the best-fitting decision-making function f_i from the data. An overview of our approach is presented in Fig. 1.

3.2. LLM traveler agents

First, we describe the design of the LLM traveler agent. This agent imitates the behavior of their human counterpart by simulating the human learning and decision-making processes defined in Eq. (1) and Eq. (3). By specifying the workflow from perception to decision, the LLM agent provides a complete implementation of $\hat{f}_i$, and is capable of being calibrated and used in simulation. The architecture of this agent is illustrated in Fig. 2.

Each agent is composed of three core elements: the LLM core, the persona, and the memory. These elements are connected with each other in the agent workflow through the agent's core functionalities. Below, we first elaborate the core elements for an LLM agent that simulates the behavior of human traveler $i \in \mathcal{N}$:

1. The LLM core (denoted by LLM): The LLM-core is the "brain" of the agent. It integrates information and, more importantly, serves as the cognitive engine for simulating the human learning and decision-making process.
2. The persona (denoted by Z_i): The persona is a natural language description of the agent's intrinsic characteristics and decision-making heuristics. It defines the agent's traits, tendencies, and factors it cares about in the decision-making process, and is a fundamental driver behind the agent's decision-making process.
3. The memory (denoted by $\mathcal{H}_i$): The memory emulates a human storing past experiences as memories. It stores a record of the agent's past decisions and their observed outcomes.

These elements are connected and operationalized in the workflow through three core functionalities: perception, memory retrieval (short-term and long-term), and decision-making. Next, we elaborate the working of each of them:

- Perception: The perception functionality manages the updating of the agent's memory $\mathcal{H}_i$. At the end of each period t, the agent receives the outcome o_i^t corresponding to the previous decision. It then incorporates this new experience into its memory by adding it to the memory set:

$$\mathcal{H}_i \leftarrow \mathcal{H}_i \cup \{(a_i^t, o_i^t)\} \tag{5}$$

  By this update, the agent maintains a complete record of its experiences. For computational efficiency, the memory is stored as a series of numerical vectors.
- Memory retrieval: When making a decision, the agent retrieves relevant experiences from its memory. To emulate the human tendency to rely on both recent events and long-term patterns (Baddeley and Hitch, 1993; Shohamy and Daw, 2015; Bornstein et al., 2017), we implement a dual-memory retrieval mechanism:

$$h_{i,s}^t = G_{i,s}(\mathcal{H}_i, t) \tag{6}$$

$$h_{i,l}^t = G_{i,l}(\mathcal{H}_i, t) \tag{7}$$

  in which $h_{i,s}^t$ and $h_{i,l}^t$ represent the retrieved short-term and long-term memories, respectively. The functions $G_{i,s}$ and $G_{i,l}$ are the corresponding time-invariant retrieval mechanisms. To ensure compatibility with the LLM, the retrieved numerical data is converted into natural language summaries by populating a structured textual template.
- Decision-making: In the decision-making stage, the agent's persona and retrieved memories are integrated into a structured prompt, q_i^t:

$$q_i^t = \text{Prompt}(Z_i, h_{i,s}^t, h_{i,l}^t) \tag{8}$$

  Following established methods (Liu et al., 2024, 2026; Sameen et al., 2025), this prompt provides the information to the LLM to simulate the traveler's action. The prompt synthesizes all influential factors into a coherent description that instructs the LLM on how to simulate the decision based on the persona, the past experiences, and the decision-making context, and the prompt template is shown in Prompt 1, Appendix A. The LLM then processes this query to generate a simulated decision, denoted by $\hat{a}_i^t$:

$$\hat{a}_i^t = \text{LLM}(q_i^t) \tag{9}$$

  In this step, the LLM, guided by the prompt, simulates the agent's cognitive process to produce the decision $\hat{a}_i^t$.

Algorithm 1 LLM traveler agent decision process in period t.

Require: Agent's persona Z_i; Memory $\mathcal{H}_i$; Retrieval functions $G_{i,s}, G_{i,l}$; an LLM core.
Ensure: Simulated action $\hat{a}_i^t$.

```
                                          ▷ Retrieve relevant experiences from memory.
h_{i,s}^t ← G_{i,s}(H_i, t)               ▷ Retrieve short-term memory.
h_{i,l}^t ← G_{i,l}(H_i, t)               ▷ Retrieve long-term memory.
                                          ▷ Construct the prompt for the LLM.
q_i^t ← Prompt(Z_i, h_{i,s}^t, h_{i,l}^t) ▷ Integrate persona and memories into a prompt.
                                          ▷ Generate the decision.
â_i^t ← LLM(q_i^t)                        ▷ Prompt the LLM to simulate the traveler's choice.
return â_i^t
```

Overall, the sequential process for an agent making a decision in time period t is presented in Algorithm 1. In our framework, we designate the memory retrieval as a modeler-specified component, while the persona is the primary learnable element. This design choice is motivated by two principles. First, it focuses the calibration process on the primary source of unobserved heterogeneity: the agent's latent persona. While the memory contains a factual history of known events, the persona is the unknown variable that requires further calibration. Second, the cognitive process of interpreting and acting upon retrieved memories is fundamentally a function of an agent's intrinsic decision-making style, which is a part of what the persona is designed to capture. By specifying a retrieval heuristic based on domain knowledge, we delegate the complex task of contextualizing and weighting this historical information to the learned persona.

3.3. LLM calibration agents

Following the definition of the LLM traveler agents in Section 3.2, the agent behavior alignment problem in (4) can be transformed into calibrating a persona Z_i for each agent to correspondingly minimize the expected future simulation error for that agent i:

$$\begin{aligned} \min_{Z_i} \quad & \mathbb{E}_{t>T}\left[\rho(\hat{a}_i^t, a_i^t) \mid \{a_i^\tau, o_i^\tau\}_{\tau=1}^T\right] \\ \text{s.t.} \quad & \hat{a}_i^t = \text{LLM}(\text{Prompt}(Z_i, h_{i,s}^t, h_{i,l}^t)) \end{aligned} \tag{10}$$

In solving problem 10, optimizing a free-form textual variable like Z_i is challenging because the search space is high-dimensional, discrete, and non-differentiable. To address this, we build the LLM calibration agent that executes an automated calibration process.

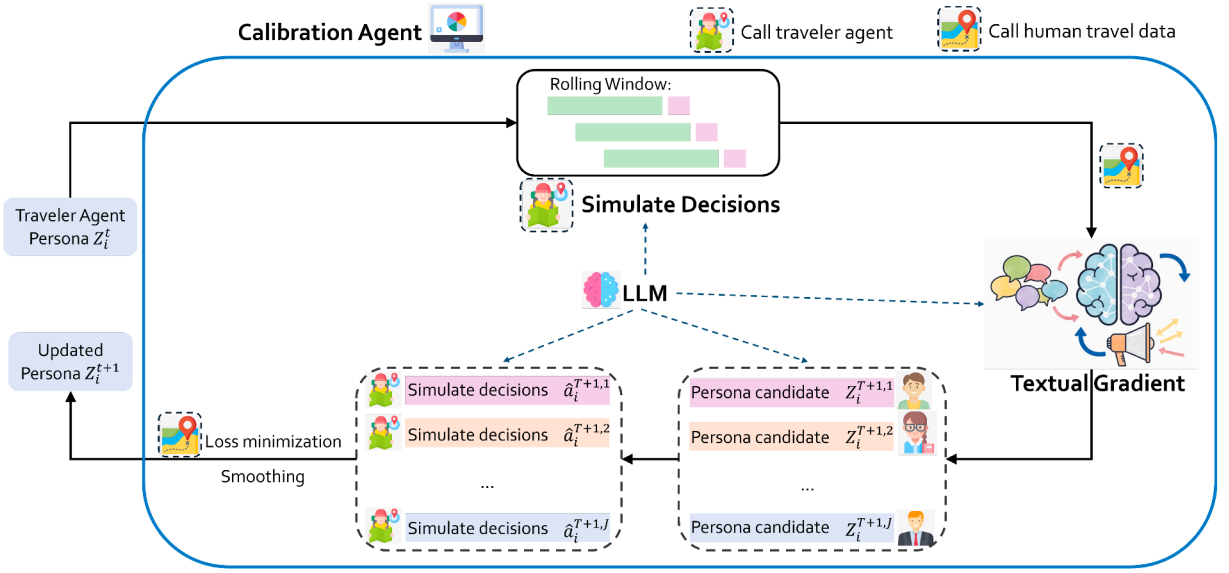


Fig. 3. Components and workflow of the LLM calibration agent.

The core idea is to leverage the reasoning capabilities of an LLM to perform a textual "pseudo-gradient" descent, inspired by similar methods in prompt optimization (Yuksekgonul et al., 2025). The overall workflow of the automatic calibration agent is depicted in Fig. 3.

The calibration process is performed iteratively at the end of each time period t when new observations of traveler behaviors and system state come to the modeler. To leverage the existing persona trained in the previous step, we do not retrain the agents' personas from scratch. Instead, we utilize a rolling update process to update the persona at every step based on the existing persona.

At the end of time t, the calibration agent focuses on a rolling window of the past t_w periods, $[t - t_w + 1, t]$, and uses data of human travelers' behavior during this window to conduct training. This approach aims to leverage the more recent observations of human travelers' behavior and prevent the number of training datapoints from growing with time at every step. Furthermore, focusing on more recent observations ensures the model adapts to the most recent thinking of the agent, which is critical for the persona to be up-to-date and able to generalize to the future. The calibration agent's workflow within each step t proceeds as follows:

1. Evaluation of the current persona. First, the calibration agent establishes a performance baseline for the current persona, Z_i^t. It simulates the agent's decisions over the active time window $[t - t_w + 1, t]$ by calling the traveler agent's decision-making function LLM:

$$\hat{a}_i^\tau = \text{LLM}\Big(\text{Prompt}(Z_i^t, h_{i,s}^\tau, h_{i,l}^\tau)\Big), \quad \forall \tau \in [t - t_w + 1, t] \tag{11}$$

2. Pseudo-gradient generation. Next, the agent generates a textual "pseudo-gradient," $\frac{\tilde{\partial}\rho}{\tilde{\partial} Z_i^t}$, for each data point for agent i in the active time window by analyzing the simulation results and attributes simulation errors to the persona. This is done by prompting the LLM core of the calibration agent (this agent functionality is denoted by LLM_g) to act as an analyst, identify the discrepancy between the simulated decision $\hat{a}_i^t$ and the human ground truth a_i^t, and suggest a textual correction to Z_i^t:

$$\frac{\tilde{\partial}\rho(\hat{a}_i^\tau, a_i^\tau)}{\tilde{\partial} Z_i^t} = LLM_g(Z_i^t, \hat{a}_i^\tau, a_i^\tau), \quad \forall \tau \in [t - t_w + 1, t] \tag{12}$$

If the traveler agent correctly simulates the decision of the corresponding human traveler ($\hat{a}_i^\tau = a_i^\tau$), the pseudo-gradient is zero (i.e., no change). If incorrect, it is a text describing the potential flaw in the persona that led to the error. We use individual data points to avoid overwhelming the LLM with long contexts, which is a known challenge for language models (Liu et al., 2023). If the predicted action is wrong, the predicted action, the ground truth action, and the decision-maker prompt are provided to the LLM via a structured prompt (shown in Prompt 2, Appendix A). The LLM then analyzes the prediction error and provides a diagnosis of the issue with the persona as its output.

3. Aggregation and candidate persona synthesis. The "pseudo-gradients" of individual observations are then synthesized to form coherent update directions of the personas. To perform the synthesis, the LLM core of the calibration agent is instructed to process the set of "pseudo-gradients" to identify common themes in their suggestion of editing of the personas, and propose J distinct candidate

update directions, $\{\tilde{\nabla} Z_i^{t,j}\}_{j=1}^{J}$. This process is denoted by LLM_i, which is detailed below:

$$\tilde{\nabla} Z_i^{t,j} = LLM_i\left(\left\{\frac{\tilde{\partial}\rho(\hat{a}_i^{\tau}, a_i^{\tau})}{\tilde{\partial} Z_i^t}\right\}_{\tau=t-t_w+1}^{t}\right) \tag{13}$$

Each update direction $\tilde{\nabla} Z_i^{t,j}$ is then used to edit the current persona Z_i^t to generate a set of J new candidate personas:

$$Z_i^{t+1,j} = LLM_e(Z_i^t, \tilde{\nabla} Z_i^{t,j}) \tag{14}$$

Operationally, each candidate persona is generated by a single query to the LLM. In the query, the diagnoses of the persona generated by the last step and the original persona are fed into the LLM using a structured prompt shown in Prompt 3, Appendix A. Responding to the query, the LLM generates a new persona by synthesizing the diagnoses and revising the original persona. The query is repeated to the same LLM with a temperature higher than 0 to generate multiple candidate personas. Here, instead of a simple mathematical aggregation of the individual pseudo-gradients, we employ a "soft" aggregation method where an integrator LLM synthesizes the feedback. We use this "soft" approach because the textual pseudo-gradients are heuristic error attributions, not true mathematical gradients, so their simple summation does not yield the optimal global update direction. Furthermore, the search space of personas is a complex semantic domain. Therefore, to facilitate a more robust, multi-directional exploration of the textual search space at each iteration, we leverage the LLM to synthesize the collection of individual feedback points into multiple distinct and plausible improvement strategies.

4. Candidate evaluation and selection. Finally, the calibration agent evaluates all candidates. For each candidate persona $Z_i^{t+1,j}$, the calibration agent calls the traveler agent to simulate the decisions over the entire window to calculate its total loss, $\mathcal{L}_i^{t+1,j}$, and compare them against the loss of the original persona, $\mathcal{L}_i^t$. The losses are calculated as follows:

$$\mathcal{L}_i^{t+1,j} = \sum_{\tau=t-t_w+1}^{t} \rho(\hat{a}_i^{\tau,j}, a_i^{\tau}) \quad j = 1, 2, \ldots, J \tag{15}$$

$$\mathcal{L}_i^{t} = \sum_{\tau=t-t_w+1}^{t} \rho(\hat{a}_i^{\tau}, a_i^{\tau}) \tag{16}$$

in which $\hat{a}_i^{\tau,j}$ are the simulated decisions of the agents obtained by using the persona $Z_i^{t+1,j}$ in a manner similar to Eq. (11). After comparing the losses of candidates, the selected persona for the agent, $\tilde{Z}_i^{t+1}$, is the best-performing candidate. If an update is triggered, a smoothing step is performed to avoid significant overfitting of personas due to randomness in short-term observations. To conduct the smoothing, a long-term baseline persona is first synthesized by an LLM (LLM_s) from a curated set of recent human traveler observations of time window length t_m, $\{a_i^{\tau}, o_i^{\tau}\}_{t-t_m+1\le\tau\le t}$. This smoothing aims to mitigate potential significant fluctuations of personas due to a relatively short time window t_w the agent is updated. Therefore, we set $t_m > t_w$ to use longer-term trends to smooth the training process. Operationally, the calibration agent uses the corresponding list of historical actions provided in the prompt (shown in Prompt 4, Appendix A to feed into the LLM core and generate a long-term reference persona. The final persona, Z_i^{t+1}, is then determined by prompting an integrator LLM (LLM_{int}) to merge the best candidate ($\tilde{Z}_i^{t+1}$) with long-term baseline. This is done by the LLM integrating the personas by the instructions provided in Prompt 5, Appendix A. Overall, the update process is:

$$Z_i^{t+1} = \begin{cases} LLM_{int}\left(\tilde{Z}_i^{t+1}, LLM_s\left(\{a_i^{\tau}, o_i^{\tau}\}_{t-t_m+1\le\tau\le t}\right)\right) & \text{if } \min_j \mathcal{L}_i^{t+1,j} < \mathcal{L}_i^t \\ Z_i^t & \text{otherwise} \end{cases} \tag{17}$$

This integrated, conditional update ensures that the agent's persona is both responsive to recent feedback and robust against transient noise, leading to more plausible long-term behavior and a more stable training process. This final update of the persona completes one full step of the calibration process.

The overall workflow of the agent is summarized in Algorithm 2.

3.4. Summary

To summarize the dual-agent framework, we provide an overview of the coupled training process involving both traveler agents and the calibration agent in Fig. 4,

The figure summarizes the full operational loop of the proposed method. After initialization from accumulated observations, the traveler agent repeatedly interacts with the environment, updates memory, and simulates decisions using its current persona. The calibration agent then compares simulated and observed actions over the rolling window, generates textual pseudo-gradients, synthesizes candidate personas, and selects an updated persona through candidate evaluation and conditional long-term smoothing. The equation references indicate how each operational step corresponds to the mathematical formulation in the methodology. Operationally, the persona-update loop begins after an initial observation period, so that each traveler has accumulated enough behavioral history to support an initial persona and the rolling calibration procedure.

Algorithm 2 LLM calibration agent: Online persona optimization at period t.

Require: Current persona Z_i^t; Human traveler data $D_t = \{a_i^\tau, o_i^\tau\}_{\tau=t-t_m+1}^{t}$; Window sizes t_w, t_m (with $t_m > t_w$); Candidate count J; LLM core.

Ensure: Updated persona for the next period, Z_i^{t+1}.

▷ **Step 1: Evaluate current persona.**

for $\tau = t - t_w + 1, \dots, t$ **do**

  $\hat{a}_i^{\tau,t} \leftarrow LLM(\text{Prompt}(Z_i^t, h_{i,s}^\tau, h_{i,l}^\tau))$ ▷ Simulate decisions with current persona.

end for

$\mathcal{L}_i^t \leftarrow \sum_{\tau=t-t_w+1}^{t} \rho(\hat{a}_i^{\tau,t}, a_i^\tau)$ ▷ Calculate initial loss on t_w window.

▷ **Step 2: Generate pseudo-gradients.**

for $\tau = t - t_w + 1, \dots, t$ **do**

  $\frac{\partial \rho_\tau}{\partial Z_i^t} \leftarrow LLM_g(Z_i^t, \hat{a}_i^{\tau,t}, a_i^\tau)$ ▷ Generate textual feedback for each data point.

end for

▷ **Step 3: Synthesize update directions.**

for $j = 1, \dots, J$ **do**

  $\tilde{\nabla} Z_i^{t,j} \leftarrow LLM_i\left(\{\frac{\partial \rho_\tau}{\partial Z_i^t}\}_{\tau=t-t_w+1}^{t}\right)$ ▷ Propose jth update strategy.

end for

▷ **Step 4: Generate and evaluate candidate personas.**

for $j = 1, \dots, J$ **do**

  $Z_i^{t+1,j} \leftarrow LLM_e(Z_i^t, \tilde{\nabla} Z_i^{t,j})$ ▷ Create jth candidate persona.

  for $\tau = t - t_w + 1, \dots, t$ **do**

    $\hat{a}_i^{\tau,t+1,j} \leftarrow LLM(\text{Prompt}(Z_i^{t+1,j}, h_{i,s}^\tau, h_{i,l}^\tau))$ ▷ Re-simulate.

  end for

  $\mathcal{L}_i^{t+1,j} \leftarrow \sum_{\tau=t-t_w+1}^{t} \rho(\hat{a}_i^{\tau,t+1,j}, a_i^\tau)$ ▷ Calculate candidate loss.

end for

▷ **Step 5: Select, smooth, and update persona.**

$j^* \leftarrow \arg\min_j \mathcal{L}_i^{t+1,j}$

if $\mathcal{L}_i^{t+1,j^*} < \mathcal{L}_i^t$ **then**

  $\tilde{Z}_i^{t+1} \leftarrow Z_i^{t+1,j^*}$ ▷ Select best-performing candidate.

  $Z_{base} \leftarrow LLM_s\left(\{a_i^\tau, o_i^\tau\}_{\tau=t-t_m+1}^{t}\right)$ ▷ Synthesize long-term baseline.

  $Z_i^{t+1} \leftarrow LLM_{int}(\tilde{Z}_i^{t+1}, Z_{base})$ ▷ Merge best candidate with baseline.

else

  $Z_i^{t+1} \leftarrow Z_i^t$ ▷ Retain original persona.

end if

return Z_i^{t+1}

4. Empirical experiment

4.1. Experiment setting

To train the LLM traveler agents and evaluate the performance of our proposed approach, we use the data from the real-world experiment conducted in Wang et al. (2025b). In the experiment, multiple groups of human participants are invited to participate in a day-to-day route choice experiment. In each experiment, a total of 15 participants are split into 2 OD groups. OD group 1 has 9 participants choosing between an "expressway" and an arterial "local1", and the remaining six participants choose between the same "expressway" and a different arterial "local2". At the beginning of each round, each traveler is presented with the total number of people choosing the "expressway" and the local arterial in the corresponding OD group, as well as their chosen route in the last round and that route's travel time. After all participants have made their choice, the travel time of each route is determined by the BPR function:

$$t = t_0\left(1 + \beta(\frac{v}{c})^4\right)$$

The specific route travel time function parameters are set up as follows:

- Expressway: $t_0 = 5$, $\beta = 0.075$, $c = 3$.
- Arterials "local1" and "local2": $t_0 = 15$, $\beta = 0.15$, $c = 5$.

in this setting, the expressway has a shorter free-flow travel time but has a smaller capacity and is more susceptible to congestion. In contrast, the local arterial has a longer free-flow travel time but has a higher capacity and impacted less by severe congestion.

The route choice game is played for 160 rounds. We select the first experiment group for testing and evaluation of our approach, and use the first 80 rounds for training the agent and the last 80 rounds for evaluation. The route choice share of each human in the

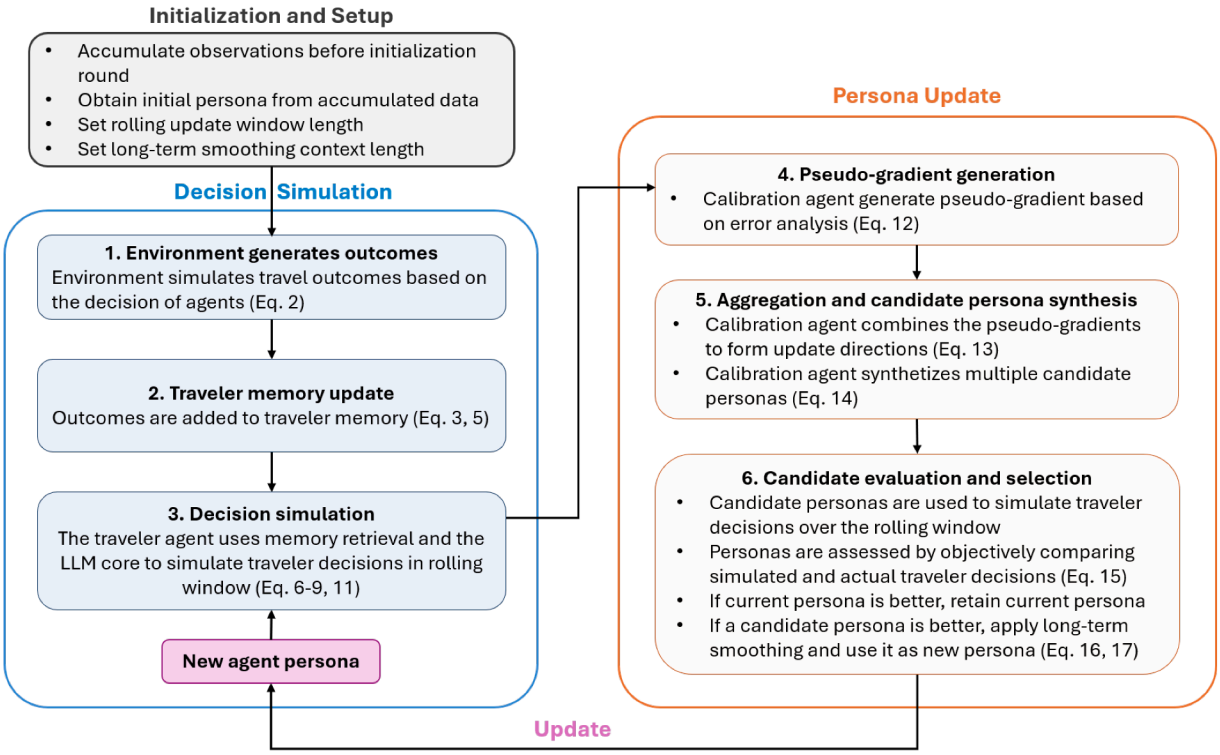


Fig. 4. Overview of the dual-agent calibration framework's algorithmic process.

Table 1
Behavior of each agent.

OD group	ID	First 20 days		Last 20 days	
		Local	Expressway	Local	Expressway
Group 1	1	37.5%	62.5%	32.5%	67.5%
	2	62.5%	37.5%	42.5%	57.5%
	3	40.0%	60.0%	37.5%	62.5%
	4	87.5%	12.5%	90.0%	10.0%
	5	38.7%	61.3%	68.7%	31.3%
	6	76.3%	23.7%	66.2%	33.7%
	7	21.3%	78.7%	28.7%	71.3%
	8	26.3%	73.7%	52.5%	47.5%
	9	48.7%	51.3%	23.7%	76.3%
	10	70.0%	30.0%	92.5%	7.5%
Group 2	11	95.0%	5.0%	98.7%	1.3%
	12	62.5%	37.5%	67.5%	32.5%
	13	36.3%	63.7%	55.0%	45.0%
	14	90.0%	10.0%	40.0%	60.0%
	15	35.0%	65.0%	62.5%	37.5%

selected dataset is shown in Table 1. For a more detailed description of the dataset, readers can refer to Wang et al. (2025b) and its corresponding supplemental materials.

As detailed in Table 1, the dataset is characterized by a rich heterogeneity in decision-making strategies. We observe a range of behaviors, from participants who maintain a consistent balance between routes (e.g., agents 1, 3, 12) to those with strong, persistent tendencies (e.g., agents 4 and 11 favor the arterial, while agent 7 favors the expressway). Crucially, the experimental design also captures non-stationary behavior, revealing several participants (e.g., agents 2, 5, 8, 13, 14, 15) who substantially alter their choice patterns over time. This diversity in both preferences and strategy evolution provides an ideal testbed for assessing a model's ability to learn proper underlying personas that can generalize to unseen conditions.

4.2. Agent specification

First, we specify the long-term and short-term memory retrieval as retrieving past memories of a certain length:

$$G_{i,s}(\mathcal{H}_i, t) = \{a_i^\tau, o_i^\tau\}_{\tau=\max(1,t-t_s)}^{t-1}$$
$$G_{i,l}(\mathcal{H}_i, t) = \{a_i^\tau, o_i^\tau\}_{\tau=\max(1,t-t_l)}^{t-1}$$

in which t_s and t_l are lengths of the recent time periods contained in the short and long-term memories, respectively. Therefore, with the indexing of memories, each retrieval task can be done relatively easily by regular programming. In the experiment, we set $t_s = 12$ and $t_l = 40$.

We use the GPT-4o API as the LLM core for both the LLM traveler agents and the LLM calibration agent. For the training process, we use a rolling window of length 8 ($t_w = 8$) at each evaluation step, and use a context size of ($t_m = 80$) for smoothing. For the calibration agent, the LLM temperature is set to 0.7, and for the traveler agent, the LLM temperature is set to 0. We initialize the training process from round 40, and the calibration agent evaluates and updates the agents' personas after every 4 rounds. The implementation of the LLM traveler decision-making prompt and the LLM calibration agent's workflow prompts are shown in Appendix A.

4.3. Comparative metrics

To compare the individual behavior alignment between the LLM traveler agent and the human traveler, we use the following metrics to quantify such a difference,

- **Accuracy**: We use the average prediction accuracy of each method in correctly predicting the human's decision, controlling for the context and past experience:

$$Accuracy = \frac{\sum_{i=1}^{15}\sum_{t=81}^{160}\mathbb{1}(\hat{a}_i^t = a_i^t)}{15 \times 80}$$

- **Weighted-average F1 score**: Since the humans in the experiment are split into two groups, each facing a different choice set for their decision at all times, we use the weighted-average F1-score to quantify the overall simulation performance. The weighted-average F1-score is calculated by:

$$F1 = \frac{3}{5}F1_1 + \frac{2}{5}F1_2$$

in which $F1_k$ is the F1-score calculated for OD-group k, calculated as:

$$F1_k = \frac{2 \times Precision_k \times Recall_k}{Precision_k + Recall_k}$$

Overall, the F1-score is calculated as the population-weighted average of the F1-score for each group, which quantifies the binary choice simulation accuracy for each OD pair in the experiment. The resulting weighted average F1-score is a value between 0 and 1, with a higher F1-score representing higher simulation performance.

- **Behavior vector**: To evaluate the alignment between simulated agents and human participants not only in decision outcomes but also in their *learning tendencies*, we adopt the classification metric proposed by Wang et al. (2025b). For each traveler i at day t, let a_i^t denote the route they choose, t_i^t denote the corresponding experienced travel time, and $\tilde{t}_i^t$ is the average travel time of the *alternative* route in the same origin–destination (OD) group. Based on whether the traveler's last trip was better or worse than the alternative and whether they switch routes, each daily decision can be categorized into four types:

$$type_i^t = \begin{cases} C^- & : \ t_i^{t-1} > \tilde{t}_i^{t-1} \text{ and } a_i^{t-1} \neq a_i^t \quad \text{(switch after loss)} \\ S^+ & : \ t_i^{t-1} > \tilde{t}_i^{t-1} \text{ and } a_i^{t-1} = a_i^t \quad \text{(stay after loss)} \\ C^+ & : \ t_i^{t-1} \leq \tilde{t}_i^{t-1} \text{ and } a_i^{t-1} \neq a_i^t \quad \text{(switch after win)} \\ S^- & : \ t_i^{t-1} \leq \tilde{t}_i^{t-1} \text{ and } a_i^{t-1} = a_i^t \quad \text{(stay after win)} \end{cases} \tag{18}$$

These four cases respectively represent a traveler's change or stay decision following a loss or win experience. Combining observations of agent behavior on each day, each agent's overall adjustment tendency over the entire experiment is summarized by a two-dimensional behavior vector:

$$\mathbf{v}_i = (C_i^-, S_i^-) \tag{19}$$

where C_i^- is the empirical probability of "switching after loss," and S_i^- is the probability of "staying after win." This vector captures both the traveler's responsiveness to negative feedback and their inertia after positive reinforcement. Based on this representation, travelers can be categorized into four behavioral archetypes: *naive* $(1,1)$, *strategic* $(0,0)$, *exploratory* $(1,0)$, and *status quo* $(0,1)$. While Wang et al. (2025b) used these discrete types for classification, we adopt a finer-grained approach by directly comparing the continuous behavior vectors of humans and LLM agents through cosine similarity:

$$\text{Sim}_{\text{behavior}} = \frac{\bar{\mathbf{v}}_{\text{LLM}} \cdot \bar{\mathbf{v}}_{\text{Human}}}{\|\bar{\mathbf{v}}_{\text{LLM}}\|_2 \times \|\bar{\mathbf{v}}_{\text{Human}}\|_2}. \tag{20}$$

in which $||\cdot||_2$ is the L^2 norm. A higher $\text{Sim}_{\text{behavior}}$ indicates closer alignment in learning and adjustment patterns.

To quantify the route flow progression difference, we use the mean absolute percentage error (MAPE) and mean squared error (MSE) between the actual and simulated route flows. The MAPE is calculated by:

$$MAPE = \frac{1}{3 \times 80} \sum_{t=81}^{160} || \frac{\hat{s}^t - s^t}{s^t} ||_1$$

in which $\hat{s}^t, s^t \in \mathbb{R}^3$ are the simulated route flow and the ground-truth route flow after time period t, correspondingly, and $|| \cdot ||_1$ is the L^1 norm. The MAPE quantifies the average simulation error of route flow over the routes for the testing period.

The MSE is quantified by:

$$MSE = \frac{1}{3 \times 80} \sum_{t=81}^{160} ||\hat{s}^t - s^t||_2^2$$

in which $|| \cdot ||_2$ is the L^2 norm. The MAPE and MSE reflect slightly different characteristics, as the latter penalizes significant deviations of the simulation from the ground truth more. Therefore, we include both metrics in our evaluation.

4.4. Compared methods

We compare our proposed approach with existing LLM-based methods and other representative theory-driven and machine learning methods in terms of performance. The baseline methods are each described below:

- Behavioral Cloning (BC) (Pomerleau, 1991): In this approach, an imitation learning method is used to directly map the system state to the traveler's decision through supervised classification. The features include a twelve-step state representation of past observations, encoding six variables: the chosen route, the number of local and expressway users, estimated travel times for both routes, and the actual travel time. A multi-layer perceptron is trained to predict the traveler's next route choice based on prior demonstrations.
- Maximum Entropy Inverse Reinforcement Learning (MaxEnt-IRL) (Ziebart et al., 2008; Zhao and Liang, 2023): In this approach, the traveler's decision-making is modeled as a Markov decision process. Using the same representation of past travel history as the BC method, this approach recovers the underlying reward function driving the traveler's choices, subject to maximum entropy regularization. The reward function is learned using a multi-layer perceptron, with future rewards discounted at a tuned factor of 0.9, allowing the model to predict decisions based on the recovered behavioral preferences.
- Base-LLM (Wang et al., 2025b): In this approach, the LLM agent retains a memory of past travel experiences, but the travel decisions are made by an unmodified LLM. At each decision-making point, the LLM is prompted with the travel context and past travel experiences before making a travel decision. The specific prompts are consistent with those of Wang et al. (2025b).
- Recursive-LLM (Wang et al., 2025a): In this approach, the LLM agent not only retains memory of past experiences, but also processes and updates their perceptions of route travel times via a recursive exponential smoothing process. For each travel decision, the LLM agents refers to their perception of route travel times as well as how many times have each route has been explored to make decisions. It uses the established day-to-day simulation paradigm but replaces the traditional agent decision-making module with a more flexible LLM agent. The specific application prompts in our assessment are consistent with those described in Wang et al. (2025a).
- Bounded-LLM: In this approach, we condition the agents with the learning model described by Jotisankasa and Polak (2006). The learning and updating of perceived travel time is only activated by a bounded rationality trigger; travelers only update their beliefs if their actual experienced travel time is surprisingly different from their perceived time, exceeding a personal threshold. The agent's route choice in the next time period is then made using a random utility maximization model, which is heavily influenced by habit; if the learning trigger was not activated, travelers are highly likely to repeat their previous choice.

5. Results

5.1. Comparison of individual-level choice behavior

First, we evaluate the accuracy of individual behavior predictions by comparing our proposed method to several baselines. Table 2 reports the average accuracy and F1-score, along with the win rates of our method versus each baseline over the individual agents.

As shown in Table 2, the proposed method demonstrates an advantage in predicting individual behavior. Among the baseline methods, BC performs the best, achieving a competitive aggregate accuracy (0.654) and F1-score (0.600). However, our approach surpasses it to achieve the highest overall accuracy (0.655) and delivers a stronger F1-score (0.635). Compared to most approaches, our proposed approach also achieves win rates over 50%. Interestingly, the theory-driven LLM agents underperformed in our experiment, as they both scored below the LLM baseline with only a memory in Wang et al. (2025b). We posit that this is because the participants' learning and adjustment behaviors deviated substantially from the perception update and bounded rational rules specified in classic theoretical models, thereby causing them to perform less effectively. This comparison also demonstrates the limitation in imposing behavioral models with *a priori* behavior assumptions in capturing human travelers' complex learning and adjustment behavior. Although the memory-only LLM baseline showed some predictive capability, likely by inferring patterns from the choice history, the superior performance of our approach validates the strategy of explicitly learning a persona. This confirms that a data-driven,

Table 2
Comparison of our model with established models on individual behavior prediction performance.

	Performance metrics		Win rate of our method on individuals	
	Accuracy	F1-score	On accuracy	On F1-score
BC	0.654	0.600	0.433	0.467
MaxEnt-IRL	0.616	0.499	0.533	0.700
Base-LLM	0.549	0.544	0.800	0.733
Recursive-LLM	0.413	0.372	0.666	0.733
Bounded-LLM	0.411	0.374	0.800	0.800
Ours	**0.655**	**0.635**	–	–

Table 3
Simulation F1-score for each agent.

OD group	ID	Ours	Base-LLM	Recursive-LLM	Bounded-LLM	BC	MaxEnt IRL
Group 1	1	0.514(5)	0.450(6)	0.657(**1**)	0.630(2)	0.620(3)	0.544(4)
	2	0.527(3)	0.453(5)	0.533(2)	0.502(4)	0.621(**1**)	0.269(6)
	3	0.557(3)	0.620(2)	0.551(4)	0.450(6)	0.666(**1**)	0.481(5)
	4	0.825(3)	0.630(4)	0.112(5)	0.018(6)	0.853(**1**)	0.853(**1**)
	5	0.685(2)	0.506(3)	0.479(4)	0.413(5)	0.745(**1**)	0.149(6)
	6	0.581(3)	0.652(2)	0.291(6)	0.306(5)	0.657(**1**)	0.528(4)
	7	0.695(3)	0.631(4)	0.731(2)	0.732(**1**)	0.593(5)	0.593(5)
	8	0.599(**1**)	0.354(2)	0.342(3)	0.334(4)	0.321(5)	0.306(6)
	9	0.582(5)	0.398(6)	0.649(3)	0.678(2)	0.685(**1**)	0.635(4)
	10	0.889(**1**)	0.871(4)	0.010(6)	0.013(5)	0.889(**1**)	0.889(**1**)
	11	0.981(**1**)	0.830(4)	0.025(5)	0.025(5)	0.981(**1**)	0.981(**1**)
Group 2	12	0.544(2)	0.569(**1**)	0.302(6)	0.344(5)	0.530(4)	0.544(2)
	13	0.520(**1**)	0.426(4)	0.428(3)	0.471(2)	0.344(5)	0.279(6)
	14	0.463(**1**)	0.292(4)	0.396(3)	0.443(2)	0.229(5)	0.229(5)
	15	0.532(2)	0.584(**1**)	0.362(4)	0.389(3)	0.268(5)	0.205(6)
Average rank		**2.40**	3.47	3.80	3.80	2.67	4.13
Average gap to #1		**0.048**	0.130	0.290	0.298	0.081	0.182

calibrated persona provides a more accurate and effective foundation for simulation, a conclusion supported by related findings in static choice contexts in transportation (Liu et al., 2024, 2026).

To gain more information on the specific performance gain of our approach, we compare the F1 score of our proposed approach with the baseline approaches in Table 3. In the table, the numbers in parentheses are the ranking of each method on the corresponding agent.

An analysis of individual agent performance, presented in Table 3, underscores the robustness of our approach. Our method is never the worst-performing model for any agent. In contrast, most baselines show significant performance fluctuations across the agent pool. Our model's performance stability is reflected in a mean rank of 2.40, which ranks first among all models, with the closest competitor being BC (2.67), and translates to a 40.7% reduction in the average performance gap to the top-ranked model (from 0.081 to 0.048). A granular analysis of per-agent performance reveals how different modeling approaches handle specific behavioral patterns. For participants with stable but mixed strategies, the theory-driven models perform well. However, their rigid assumptions fail when applied to agents who exhibit strong route biases (e.g., 4, 10, 11). The Base-LLM shows the opposite weakness: while sometimes competitive, it often fails to accurately model agents with balanced choice patterns (e.g., 1, 2, 9, 13). For agents who exhibit strong biases for routes, the BC and MaxEnt-IRL baselines perform well, often tying for the top score by effectively memorizing the dominant choice. Yet, they struggle when agents display balanced or shifting choice patterns (e.g., agents 8, 13, 14, 15), and our approach performs well in these cases. Overall, our proposed approach distinguishes itself by achieving competitiveness across all of these behavioral types. Its success with agents exhibiting shifting patterns provides evidence that our framework successfully learns the evolution of an agent's strategy, a critical capability for realistic modeling and simulation.

The learned personas of agents reflect their decision-making tendencies, and the differences in each agent can also be reflected in the learned personas. Some learned personas of representative agents are shown in Table 4:

Table 4 shows the learned personas of four representative agents: agents 3 and 6, who maintain a relatively balanced distribution between the two routes, and agents 7 and 11, whose choices are heavily inclined towards one option. The learned personas effectively capture these distinct tendencies by articulating the agents' decision-making priorities. For agents 3 and 6, their personas reflect a careful consideration of both short- and long-term memory to evaluate route performance and reliability. Their decision-making prioritizes a rational evaluation of travel time and is not biased towards one particular route. These underlying mechanisms drive them to react to changing conditions and switch routes strategically to minimize their travel times. In contrast, the learned personas of agents 7 and 11 reveal a strong cognitive bias to default to one route. Agent 7's persona states it will "Switch to local1 only if it demonstrates sustained improvement over 5 consecutive trips," while agent 11's contains the belief that 'local2' is the safer, more

Table 4
Example of learned personas of agents.

ID	Learned persona
6	**Short-term processing rule**:Prioritize recent experiences by evaluating the travel times from the last 5 trips. Favor routes that consistently deliver shorter travel times, but avoid routes that have shown extreme or unpredictable delays in the recent past (e.g., significantly longer durations compared to the average). **Long-term strategy rule**: Maintain a default bias toward routes that have historically provided the most consistent and reasonable travel times. Be cautious of routes with a history of frequent extreme delays, even if they occasionally offer faster times. **Decision Heuristic**: Combine short-term and long-term rules by weighting recent trip performance more heavily when making immediate decisions, while adjusting for historical patterns of reliability. Choose the route that balances shorter recent travel times with long-term consistency, avoiding routes with high variability or risk of significant delays.
3	**Short-term processing rule:** Monitor the last 3–5 trips on the current route. Switch to the alternative route if delays exceed 25 min in two consecutive trips *or* variability exceeds 15 min across three trips, unless recovery trends are evident (e.g., two consecutive trips with delays below 20 min or consistent improvement over three trips). Treat delays exceeding 50 min as critical signals of instability, but avoid immediate reactions unless instability persists across multiple trips. Favor routes showing travel times consistently under 20 min, but temper this preference by considering one-off shorter travel times (e.g., under 20 min) as indicators only if aligned with broader stability across recent trips. **Long-term strategy rule:** Maintain preference for routes with historically faster average travel times and lower variability, while penalizing recurring extreme delays (>50 min) observed in at least three trips within a recent 10-trip window. Reward recovery trends when recent performance demonstrates stabilization, and prioritize routes showing consistent travel times and low variability over longer periods. Avoid reverting to historically favorable routes unless recent data confirms reliability and consistency. Default biases toward expressways or other historically faster routes should be moderated by higher local distribution trends and consistent performance, ensuring alignment with broader reliability indicators. **Decision Heuristic:** Combine short-term and long-term rules with dynamic weighting: assign 60% weight to long-term trends and 40% weight to short-term data, increasing short-term weight only when recovery trends are clear or instability persists (e.g., sustained delays or variability over three or more trips). Ensure transient disruptions or isolated events do not disproportionately influence decisions, while factoring in significantly shorter travel times (e.g., under 20 min) as secondary indicators only if consistent with broader trends. Integrate traffic distribution trends, favoring routes with higher commuter traffic only when aligned with stable and consistent performance.
7	**Short-term processing rule**: Evaluate the average travel time across the last 5 trips, prioritizing routes with consistently lower averages. Avoid overreacting to isolated deviations unless they persist for at least 3 consecutive trips. Incorporate the travel time of the most recent trip as supplementary information by assigning a weight: negative if greater than 20 min, positive if less than 15 min, and neutral if between 15 and 20 min. Use these weights to refine the broader pattern evaluation. **Long-term strategy rule**: Maintain a default bias toward the expressway, as historical data confirms its faster travel times under favorable conditions. Define "consistent poor performance" as the expressway exceeding a specific threshold (e.g., 20 min) for 5 consecutive trips or averaging over 30 min in recent consecutive trips. Switch to local1 only if it demonstrates sustained improvement over 5 consecutive trips with an average travel time significantly lower than the expressway. Revert to the expressway once conditions stabilize or variability decreases. **Decision heuristic**: Balance short-term trends and long-term data by prioritizing the expressway when trends are mixed or inconclusive. Switch to local1 only if both short-term averages and long-term data clearly indicate persistent poor performance on the expressway and sustained improvement on local1. If the short-term weight for the most recent trip is negative and aligns with broader data showing consecutive poor expressway performance, prefer local1 for the next trip. Recent data should only outweigh long-term trends if it shows sustained, extreme deviation over at least 5 consecutive trips, ensuring optimal travel time while minimizing risk.
11	**Short-term processing rule**: Prioritize the most recent trips by evaluating the travel time of the last few choices. If my most recent route consistently results in travel times close to or better than my historical average (around 15.9 min for 'local2'), I should favor repeating that choice. If my latest choice deviates significantly or results in worse outcomes, reconsider the alternative. **Long-term strategy rule**: Avoid high-risk options based on past extreme negative experiences. The 'expressway' route is inherently biased as unreliable due to historical outliers (e.g., 72.8 min), even if it occasionally offers competitive travel times. My default belief is that 'local2' is the safer, more consistent option, with predictable and moderate travel times. **Decision heuristic**: Combine short-term validation with long-term bias to make a final choice. If the short-term travel times for 'local2' remain consistent and below historical risk thresholds (e.g., avoiding spikes like 17.2 min or worse), I should choose 'local2.' Only consider 'expressway' if recent data strongly suggests a significant and reliable improvement in travel time distribution without extreme risks.

consistent option." While both agents will deviate from their default choice when evidence is overwhelmingly clear, their personas enforce a high threshold for switching and a strong inclination to revert to their preferred option. Overall, the analysis confirms that our framework generates good and interpretable models, successfully capturing a spectrum of learning and decision-making styles in travelers' day-to-day behavior.

Also, as evidenced in Table 3, our dual-agent approach can learn the agent's gradual update of their personas as they interact with the system and obtain more experiences, therefore helping out-of-sample generalization. This process is also evident in the agent's persona change in the learning process, as we show by using the example of agent 8 in Table 5.

The evolution of agent 8's persona in Table 5 provides a clear longitudinal example of the calibration process. The agent begins with a naive and highly reactive strategy, heavily weighing the most recent trips and over-penalizing singular extreme delays, all while relying on a simple default bias for the expressway. Through the training process, this persona matures into a significantly more sophisticated and robust policy. The final persona explicitly demonstrates that the agent has learned to differentiate signal from noise: it now prioritizes "sustained patterns" and "long-term reliability" while actively "avoiding overreaction to anomalies" and "isolated negative events". Consequently, the agent's behavior evolves from heavily favoring the expressway in the first 20 days to having balanced choices between routes in the last 20 days. This progression of persona provides an example of the training process in action, validating our adaptive learning mechanism.

Table 5
The starting and ending personas of agent 8.

Time	Learned persona
Start of training	**Short-Term Processing Rule**: Prioritize recent experiences by heavily weighing the travel times from the last 5 trips. If a route consistently performs better in terms of shorter travel time within this window, favor that route. However, penalize routes that result in significantly higher-than-average travel times (e.g., exceeding 50 min), even if they performed well during other recent trips. **Long-Term Strategy Rule**: Default to believing the expressway is generally faster and more reliable, unless historical data shows repeated extreme delays (e.g., over 70 min). Assume that lower travel time is more likely on the expressway, especially when the distribution favors expressway usage over local routes. **Decision Heuristic**: Combine short-term and long-term rules by first evaluating the recent performance (short-term rule) and then adjusting the decision based on long-term bias. Choose the expressway if recent travel times are comparable or better than local routes, but switch to local routes if the expressway has shown consistently poor performance or if recent data shows a significant risk of long delays. Prioritize avoiding extreme delays over adherence to historical bias.
End of training	**Short-Term Processing Rule**: Evaluate the last 5 trips with a focus on identifying sustained patterns rather than isolated events. Prioritize routes with short-term stability and consistently moderate travel times, rewarding reliable performance even if occasional variability is present, provided extreme delays are infrequent and do not form clear worsening trends. Avoid penalizing routes for isolated negative events unless extreme delays occur repeatedly over at least 3 consecutive trips. Ensure negative patterns are assessed with proportional penalties, emphasizing overall consistency and avoiding overreaction to anomalies. **Long-Term Strategy Rule**: Favor routes with faster historical averages and sustained reliability, maintaining a strong bias toward options that consistently demonstrate long-term superiority. Penalize extreme delays (>50 min) only if they occur repeatedly and disrupt long-term reliability. Balance flexibility to shift toward alternatives when sustained improvements emerge with maintaining preference for historically faster options. Minimize short-term disruptions from isolated events unless they clearly impact long-term reliability or show sustained worsening patterns. **Decision Heuristic**: Combine short-term trends and long-term reliability, ensuring decisions emphasize sustained advantages across both dimensions. Prioritize routes that demonstrate consistent reliability and faster historical averages while ensuring recent performance reflects broader patterns rather than isolated anomalies. Slightly favor long-term superiority when historical trends clearly outweigh short-term disruptions. Apply penalties proportionally, balancing rewards for sustained reliability and moderate travel times across both short-term and long-term considerations without overreacting to isolated negative events.

Table 6
Cosine similarity of simulated and human behavior vectors for each agent.

OD group	ID	Ours	Base-LLM	Recursive-LLM	Bounded-LLM	BC	MaxEnt IRL
Group 1	1	0.880(5)	0.921(4)	0.942(3)	0.968(2)	1.000(**1**)	0.796(6)
	2	0.948(4)	0.995(**1**)	0.989(2)	0.986(3)	0.922(5)	0.836(6)
	3	0.980(4)	0.938(5)	0.998(**1**)	0.983(3)	0.987(2)	0.934(6)
	4	1.000(**1**)	0.799(6)	0.993(3)	0.996(2)	0.979(4)	0.979(4)
	5	0.990(2)	0.887(4)	0.777(6)	0.817(5)	0.990(2)	0.996(**1**)
	6	1.000(**1**)	1.000(**1**)	0.998(4)	1.000(**1**)	0.976(5)	0.956(6)
	7	0.851(4)	0.999(**1**)	0.929(2)	0.916(3)	0.756(5)	0.756(5)
	8	0.999(2)	0.940(6)	1.000(**1**)	0.999(2)	0.979(4)	0.948(5)
	9	0.999(3)	0.987(6)	0.999(3)	0.999(3)	1.000(**1**)	1.000(**1**)
	10	1.000(**1**)	0.977(6)	1.000(**1**)	1.000(**1**)	0.988(4)	0.988(4)
Group 2	11	1.000(**1**)	0.949(6)	1.000(**1**)	0.999(5)	1.000(**1**)	1.000(**1**)
	12	0.977(3)	0.993(2)	0.943(4)	0.927(6)	0.997(**1**)	0.935(5)
	13	0.998(**1**)	0.944(5)	0.984(3)	0.993(2)	0.981(4)	0.877(6)
	14	0.997(2)	0.999(**1**)	0.958(6)	0.972(3)	0.966(4)	0.966(4)
	15	0.988(2)	0.848(6)	0.930(5)	0.990(**1**)	0.965(3)	0.947(4)
Average rank		**2.40**	4.00	3.00	2.80	3.07	4.27
Average value		**0.974**	0.945	0.963	0.970	0.966	0.928

We first present a quantitative evaluation of behavioral fidelity in Table 6. This table reports the cosine similarity between the simulated behavior vector (C^-, S^-) and the human ground truth for all 15 agents. The number in parentheses indicates the performance rank of each method for that specific agent, from (1) for best (highest similarity) to (6) for worst.

The summary statistics at the bottom confirm the superior performance of our framework. Our method achieves the best average rank (2.40) and the highest average value (0.974), indicating it most consistently and accurately replicates the true human behavioral vectors. The Bounded-LLM is the closest competitor (average rank 2.80, average value 0.970), while the other baselines perform worse on both metrics. Notably, our method achieves a perfect similarity score of 1.000 for four different agents (agents 4, 6, 10, and 11) and in some other cases near-perfect scores (agents 8, 9, 13, 14), demonstrating good alignment in numerous cases. While this quantitative data confirms that our method is the most accurate, the following four-corner analysis will visually explain why by decomposing these vectors.

Fig. 5 visualizes the global distribution of these vectors (days 20–39) across methods for the ground truth against the tested methods. The plots reveal a clear distinction. For our framework (purple circles), the markers have mostly dark colors (indicating values ≈ 1.0), showing high cosine similarity with humans, which confirms this strong quantitative alignment. This suggests that our dual-agent architecture effectively models the human separation between learning and behavioral adjustment. In contrast, the baseline methods perform less well. The Base-LLM (green diamonds) agents are erratically distributed and exhibit low cosine similarity. While

Table 7
Model comparison for aggregate route flow simulation under controlled system dynamics.

	MAPE	MSE
BC	26.3%	2.49
MaxEnt-IRL	28.0%	3.03
Base-LLM	42.3%	7.49
Recursive-LLM	81.3%	22.36
Bounded-LLM	78.9%	21.24
Ours	40.0%	5.94

the Recursive-LLM (blue triangles) and Bounded-LLM (purple inverted triangles) agents correctly learn the high S^- component, they fundamentally misunderstand inertia. Both incorrectly exhibit much higher C^- values. The BC method (orange hexagons) overall underestimates the S^- component of the agents, and the MaxEnt-IRL approach (dark red hexagons) is biased toward the status quo extreme of decision-making. In conclusion, Fig. 5 demonstrates that baseline LLM approaches tend toward an overly reactive "lose-shift" behavior. Our framework is the only one to uniquely model human behavioral inertia, achieving superior alignment with the human travelers.

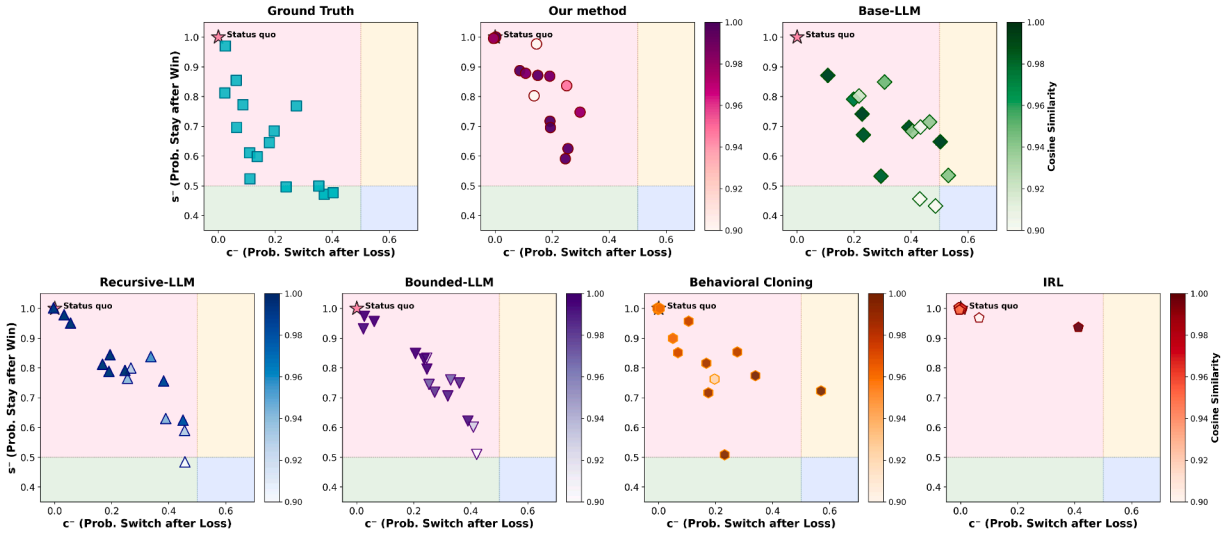


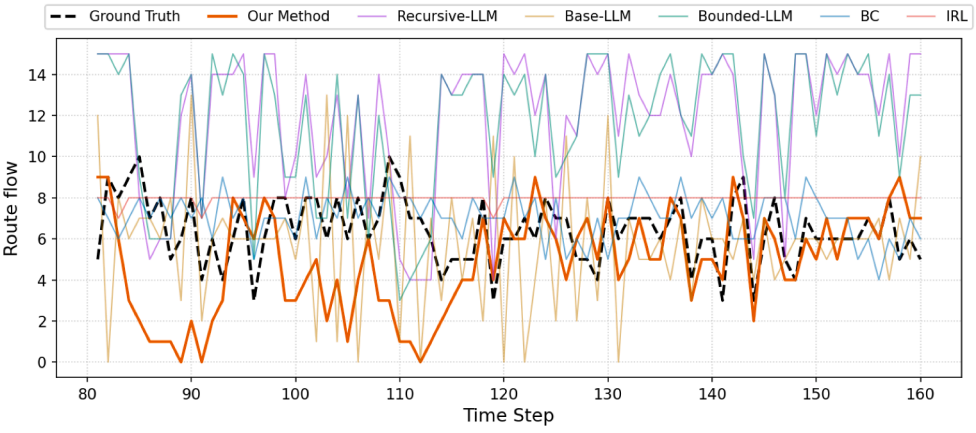
Fig. 5. Global behavior vector distribution (Day 20–39) across methods. Marker color intensity reflects cosine similarity to the corresponding human traveler.

5.2. Comparison of system dynamics

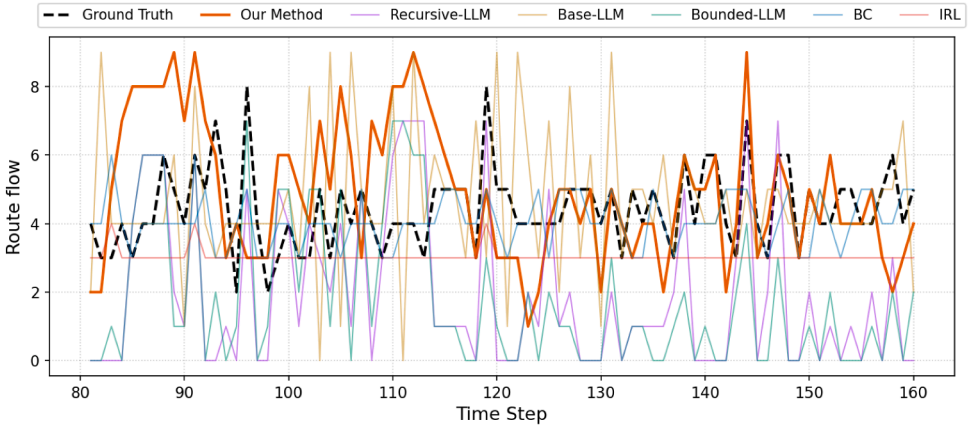
In this section, we compare the aggregate simulation performance of our methods and established approaches over the simulated route flow. First, we control the system dynamics by providing all models with the ground truth system state from the previous days, and simulate the route flow progression for the last 80 days. This approach is designed to test the overall route choice tendencies of the models while isolating their aggregate behavior from the confounding effects of compounding simulation errors. The simulation results are presented in Fig. 6, while evaluations of the simulation results are presented in Table 7.

The simulation results, presented in Fig. 6, demonstrate our model's ability to accurately reproduce system-level route flows. Qualitatively, our model's simulated flow not only converges with the ground truth, particularly in the latter half of the simulation (rounds 120+), but also mirrors its characteristic smoothness. This contrasts sharply with the Base-LLM, which produces a highly volatile flow, and the other two baselines, which fail to capture the correct trend. This suggests their simpler decision mechanisms or recursive structures are not well-suited for leveraging accurate state information, potentially leading to over-reactions and chaotic system behavior. On the contrary, the learning methods' predicted route flows are less volatile than the LLM-based methods and the human ground truth. The MaxEnt-IRL method, in particular, produces highly stable predictions, indicating that it struggles to capture the dynamic, day-to-day behavioral adaptation of the human population.

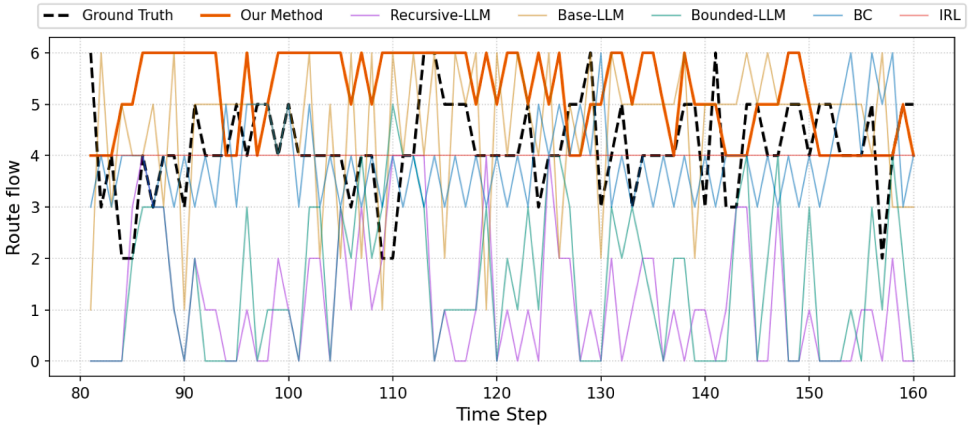
Interestingly, the results in Table 7 show that the classical machine learning baselines achieve very strong aggregate flow predictions. BC and MaxEnt-IRL attain MAPE values of 26.3% and 28.0%, respectively, outperforming the LLM-based baselines and our primary implementation. An explanation for this is that because these ML models are trained end-to-end on numerical features ex-



(a) Expressway

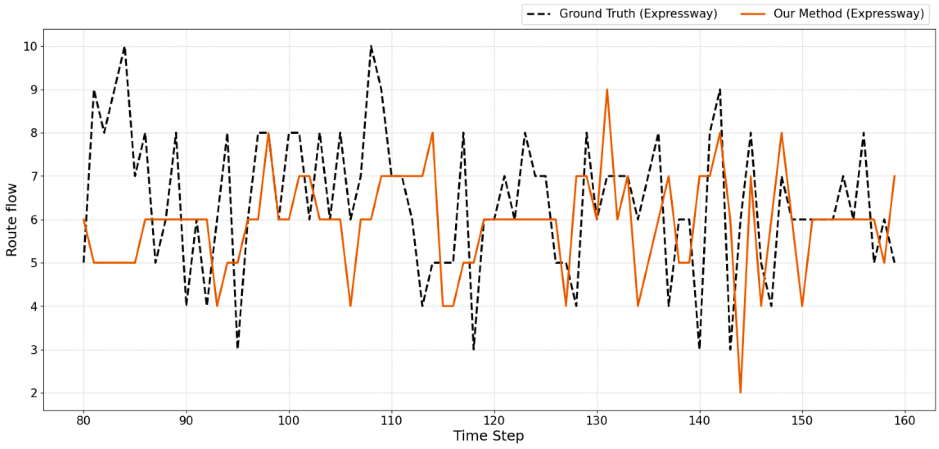


(b) Arterial "local 1"

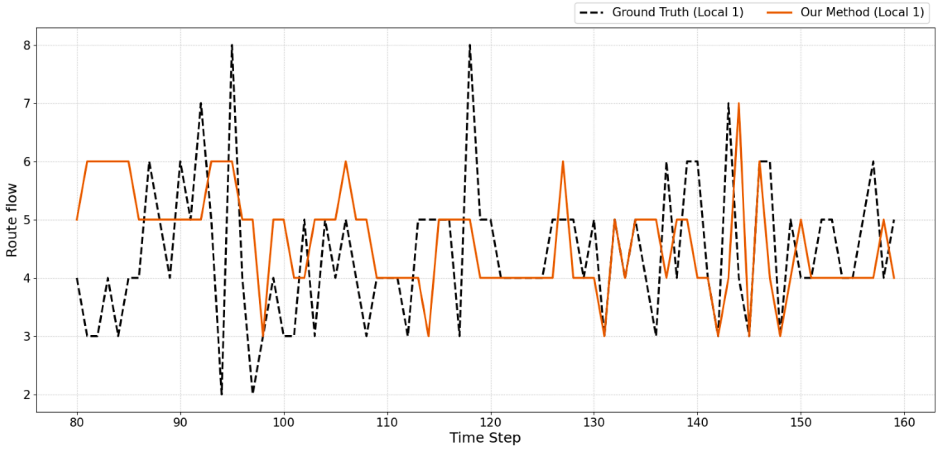


(c) Arterial "local 2"

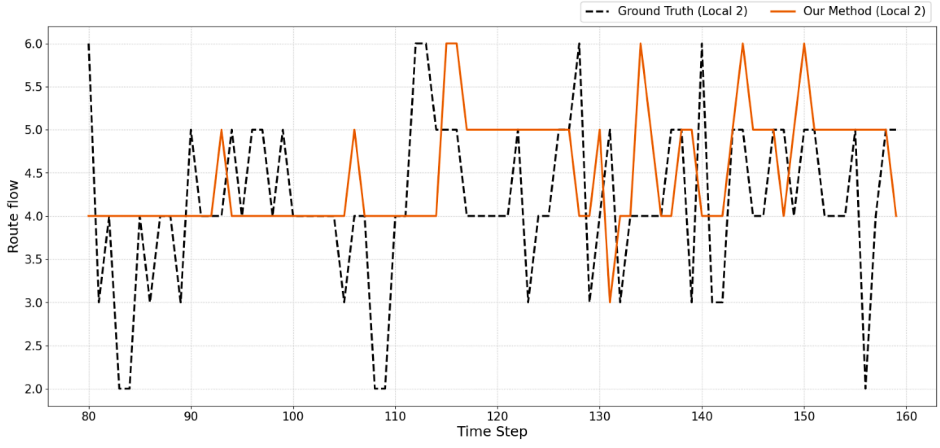
Fig. 6. Simulated route flows under controlled system dynamics.



(a) Expressway



(b) Arterial "local 1"



(c) Arterial "local 2"

Fig. 7. Simulated route flows under uncontrolled system dynamics.

Table 8
Performance of the proposed framework with different LLM backbones.

LLM	Individual behavior prediction		Route flow prediction	
backbone	Accuracy	F1-score	MAPE	MSE
GPT-4o	0.655	0.635	40.0%	5.94
Gemini-3.0-flash-preview	0.640	0.638	31.5%	2.25
Claude-Haiku-4.5	0.721	0.728	25.7%	2.13

plicitly to minimize prediction error, they are inherently optimized for reproducing aggregate distributional patterns. However, our proposed dual-agent framework is entirely model-agnostic. When evaluated against other LLM-driven models, our approach already achieves the highest aggregate accuracy, successfully avoiding the chaotic volatility of the Base-LLM and the severe over-reactions of the theory-driven Recursive and Bounded agents.

Next, we evaluate the performance of our approach by conducting an experiment that removes the controls in Fig. 6 to resemble standard counterfactual simulations. In this setting, agents learn and adapt based solely on the travel times they experience, processed through their memory mechanism, without access to the ground truth flows from the previous step. This represents a more realistic scenario characterized by imperfect information and experiential learning.

Fig. 7 provides an isolated comparison between the route flows generated by our model under uncontrolled system dynamics (orange line) and the ground truth (dashed black line). The simulation successfully captures the qualitative trends and persistent fluctuations of the ground truth, demonstrating effective adaptive learning. However, the simulated flows exhibit occasional lags or deviations. This is likely because the agents are learning from inherently noisy system dynamics and face delays in reacting to changes, as they rely solely on experience rather than perfect state information. Despite this reduced moment-to-moment accuracy, the model's ability to reproduce these key dynamic characteristics validates its capacity to approximate system behavior using only experiential feedback.

5.3. Cost and robustness analysis

To evaluate our framework's robustness to other LLM backbones, we repeat our experiment using the same data, setting, and prompt with two alternative LLMs from different providers: Gemini-3.0-flash-preview and Claude-Haiku-4.5. The resulting individual behavior prediction performance and the route flow prediction performance are shown in Table 8,

As shown in Table 8, the proposed framework remains effective across all three tested backbones. On individual behavior prediction, Gemini achieves performance comparable to GPT-4o, while Claude further improves both accuracy and F1-score. On route-flow prediction, both Gemini and Claude also improve over the GPT-4o implementation, achieving lower MAPE and MSE. Moreover, the Gemini and Claude variants both outperform the baseline methods in aggregate flow distribution, indicating that the proposed dual-agent framework is not tied to a single LLM backbone and can remain competitive or stronger when instantiated with alternative models.

Next, we report the computational cost and latency of our experiment. On average, a single calibration update for one agent requires about 40 API calls, totaling approximately 51,000 input tokens and 2000 output tokens, with an end-to-end latency of 33 s. In contrast, one traveler decision simulation step requires only a single API call with about 1370 input tokens, 3 output tokens, and 0.2 s of latency. Across the full experiment, the average runtime per agent is approximately 6 min. The corresponding per-agent monetary cost depends on the LLM API, and is approximately USD 1.75 for GPT-4o, USD 0.37 for Gemini-3.0-flash-preview, and USD 0.72 for Claude-Haiku-4.5. These results indicate that the current implementation is computationally manageable for controlled offline experiments, while also making clear that the calibration stage is the dominant source of cost.

The framework's computational overhead scales with the number of agents, calibration iterations, persona search space, prompt length, and underlying LLM inference costs. To enhance scalability, future large-scale applications could optimize efficiency by calibrating representative agents for broader traveler segments, implementing adaptive stopping criteria to minimize redundant updates, and narrowing the candidate pool during calibration. Furthermore, computational optimization techniques such as prompt batching, caching, and parallelization can also be implemented to reduce processing time. Based on the experimental results, our framework is suited to offline planning settings characterized by complex and heterogeneous traveler behavior. By providing interpretable insights into how individuals learn and adapt over time, it can serve as a behavioral modeling layer for discovering representative personas and informing simulations.

6. Conclusion

In this paper, we introduce a novel dual-agent framework to automatically align LLM agents with human travelers for simulating the dynamic learning and adjustment of humans in transportation systems. Our approach features a set of LLM traveler agents, which use a memory system and a learnable persona to simulate the learning and subsequent decision-making of human travelers, and a supervisory LLM calibration agent which performs an automated, online optimization of the traveler's persona using a textual "pseudo-gradient" descent method. Using an online automatic simulation and calibration procedure, we transform the complex task of behavioral alignment into an iterative, rolling-window optimization problem that is jointly solved by the simulations of LLM traveler agents and the learning updates of the LLM calibration agent. Our empirical evaluation on a real-world day-to-day route choice dataset

demonstrates that our framework significantly outperforms existing baselines in individual learning accuracy and also performs better than existing LLM agents in aggregate flow prediction accuracy. Furthermore, our analysis of the model performance on individual travelers reveals the method's ability to align not just with behavioral outcomes, but with the underlying decision-making tendencies and their evolution. We observe that our model significantly outperforms baseline models when the traveler exhibits significant changes in their decision-making strategies with time, and the assessment of the learned personas reveals that our approach can explicitly capture such changes in persona. Furthermore, using a decision-making tendency evaluation in Wang et al. (2025b), our model also achieves the best alignment with the human ground truth, further strengthening the alignment. Computationally, we reduce the number of tokens needed for the training and evaluation process by summarizing all learned behavior into a compact persona and a fixed-length memory for the traveler agents, and use a small rolling window for the calibration agents. This strategy reduces computational overhead, making the alignment process more cost-effective.

Future research can build on this study by focusing on several promising directions. First, enhancing the computational efficiency of the online calibration process is a key area for development. This could involve exploring more token-efficient methods for aggregating pseudo-gradients, such as developing single-pass "batch" updates, or refining the candidate generation strategy to reduce the number of LLM queries required per update, thereby lowering computational costs and latency. Second, future work could explore the joint learning of both the persona and the memory retrieval mechanisms. While we fixed these retrieval heuristics for identifiability and learning stability, allowing the agent to also learn its own information-weighting and memory utilization strategies could create agents with even more adaptive and personalized decision-making styles. Another direction of future work is to extend the empirical evaluation of the framework to richer decision settings, such as multi-route, joint route–departure-time, or multi-attribute choice problems, which would help assess how well the proposed calibration mechanism transfers to more complex behavioral environments. Finally, a direction for future work is to expand the behavioral validation of the learned personas using additional decision settings and complementary forms of behavioral evidence. Such extensions would further strengthen the interpretability of the framework and its ability to characterize human decision patterns.

CRediT authorship contribution statement

Tianming Liu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization; **Jirong Yang:** Writing – original draft, Visualization, Validation, Methodology, Investigation; **Yafeng Yin:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization; **Manzi Li:** Writing – original draft, Visualization, Validation, Methodology; **Linghao Wang:** Methodology, Data curation; **Zheng Zhu:** Methodology, Data curation, Conceptualization.

Acknowledgements

The work described in this paper was partly supported by research grants from the National Science Foundation (CMMI-2233057 and 2240981).

Data availability

Data will be made available on request.

Appendix A. Prompts

Prompt 1: Simulate traveler decision

```
You are participating in a commuting simulation with 15 people in total. The participants are split into two groups:
- Group OD1 (9 people): Can choose between 'expressway' and 'local1'.
- Group OD2 (6 people): Can choose between 'expressway' and 'local2'.
You are a commuter agent. Your personality and decision-making rules are defined below. Apply these rules to the recent memory provided in the user prompt to make the best choice. — YOUR PERSONA (RULES) —self.dynamic_prompt.prompt
Here is your recent memory: — YOUR MEMORY (RECENT HISTORY) —memory_context
Now it is step step.step_in_day of day day. Based on your persona and this memory, what route will you choose next?
Options: expressway or local_name.
Only output the single option word.
```

Prompt 2: Generating individual observation error diagnosis

```
You are a behavioral analyst debugging an agent's persona.
An agent using the ruleset (persona) below made a wrong decision when presented with its memory.
Analyze the rules to identify the flaw. Why did its ruleset lead it astray in this specific situation?
- **Full Prompt (Rules + Memory) That Failed:**failed_prompt
- **Agent's Incorrect Choice:** pred
```

- **Human's Correct Choice:** step.route_choice
Based on this, pinpoint the flaw in the agent's RULES for each dimension:
- **Flaw in Short-Term Processing Rule:** …
- **Flaw in Long-Term Strategy Rule:** …
- **Flaw in Decision Heuristic:** …

Prompt 3: Generating persona updates based on collection of error analysis

You are an expert at refining an agent's decision-making rules. Below is the agent's original set of rules (persona) and a collection of analyses detailing where its rules led to wrong decisions.
Your task is to synthesize these failure analyses to find the most common flaws in the agent's logic. Based on these recurring flaws, rewrite the agent's ruleset to make it a better decision-maker. **Crucially, do not try to fix every single error.** Focus on upgrading the core rules to address the most significant, repeated flaws.Maintain the 'Short-Term Processing Rule', 'Long-Term Strategy Rule', and 'Decision Heuristic' structure in your revised persona. Write the new persona in an active voice, as if you are giving the agent new, improved instructions.
Original Persona (Rules):self.prompt
Collection of Failure Analyses:feedbacks
Your final output must contain ONLY the three revised persona rules under their respective headings. Do not include any extra explanation, summary, or introductory text.

Prompt 4: Generating of the smoothing persona

You are an AI psychologist defining a commuter's decision-making model based on their past behavior. Create a set of rules (a persona) that explains HOW this person thinks.
Based on the following historical experience, define the agent's persona. Your description must explicitly address these three points as active rules:
1. **Short-Term Processing Rule:** How should I react to my most recent trips?
2. **Long-Term Strategy Rule:** What is my default belief or bias?
3. **Decision Heuristic:** How do I combine the short-term and long-term rules to make a final choice?
"Historical experience:memory_context
Write this persona as a set of direct instructions. Your final output must contain ONLY the three persona rules under their respective headings.

Prompt 5: Integration of the smoothing persona

You are an AI psychologist refining an agent's strategy.
You are refining an agent's decision-making model. Below are two descriptions of its strategy:
1. **The Current Refined Persona:** This is the agent's working set of rules, including specific lessons learned from recent mistakes. This should be the foundation.
2. **The 20-Day Trend Baseline:** This persona represents general trends observed over the last 20 days.
Your task is to **synthesize** these into an updated persona. Start with the **Current Refined Persona** as the base. Then, review the **20-Day Trend Baseline** and use its insights *only* to moderate or temper any rules in the refined persona that seem overly reactive or potentially overfit to the very latest data. The goal is a balanced persona that prioritizes learned lessons but smooths out extreme reactions based on medium-term trends. Do not simply replace the refined rules with the baseline trends.
— Current Refined Persona (Base for Synthesis) —refined_persona
— 20-Day Trend Baseline (Moderating Insights) —strategic_baseline_20day
Your final output must contain ONLY the new, synthesized persona, following the three-rule structure. Do not include any extra explanation or summary

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